

Pre-Lab Activity

Pre-Lab 4 - Part 1

Updated turtle_client.py:

```
from action_msgs.msg import GoalStatus
from turtle_interfaces.action import MakeSquare

import rclpy
from rclpy.action import ActionClient
from rclpy.node import Node
from rcl_interfaces.msg import ParameterDescriptor # <-- descriptor import

class TurtleSquareClient(Node):
    def __init__(self):
        super().__init__('square_client')
        self._action_client = ActionClient(self, MakeSquare, 'make_square')

        # Declare parameter with descriptor (double)
        self.declare_parameter(
            'square_size',
            50.0,
            ParameterDescriptor(description='Side length of the square (pixels)')
        )

    def goal_response_callback(self, future):
        goal_handle = future.result()
        if not goal_handle.accepted:
            self.get_logger().info('Goal rejected')
            return
        self.get_logger().info('Goal accepted')
        self._get_result_future = goal_handle.get_result_async()
        self._get_result_future.add_done_callback(self.get_result_callback)

    def feedback_callback(self, feedback):
        self.get_logger().info(f'Received feedback: {feedback.feedback.current_pose}')

    def get_result_callback(self, future):
        result = future.result().result
        status = future.result().status
        if status == GoalStatus.STATUS_SUCCEEDED:
            self.get_logger().info(f'Goal succeeded! Result: {result.final_pose}')
        else:
            self.get_logger().info(f'Goal failed with status: {status}')
        rclpy.shutdown()

    def send_goal(self):
        self.get_logger().info('Waiting for action server...')
        self._action_client.wait_for_server()

        # Read the parameter (double_value)
        size = self.get_parameter('square_size').get_parameter_value().double_value

        goal_msg = MakeSquare.Goal()
        goal_msg.square_size = float(size) # <-- set square_size from parameter

        self.get_logger().info(f'Sending goal request with square_size={size}...')
        self._send_goal_future = self._action_client.send_goal_async(
            goal_msg,
            feedback_callback=self.feedback_callback
        )
        self._send_goal_future.add_done_callback(self.goal_response_callback)

def main(args=None):
    rclpy.init(args=args)
    action_client = TurtleSquareClient()
    action_client.send_goal()
    rclpy.spin(action_client)

if __name__ == '__main__':
    main()
```

Test default execution:

```
yahboom@yahboom-virtual-machine: ~  
yahboom@yahboom-virtual-machine:~$ source /opt/ros/humble/setup.bash  
yahboom@yahboom-virtual-machine:~$ source ~/S26_RobotProgramming_Eloquent_Elusor/Lab4/wojnae/roscourse_ws/install/setup.bash  
yahboom@yahboom-virtual-machine:~$ ros2 run turtle_test turtle  
[INFO] [1773117811.502597234] [turtle_square_server]: MakeSquare action server ready.  
[INFO] [1773117837.247764716] [turtle_square_server]: Received goal: square_size=50.0  
[INFO] [1773117837.250685272] [turtle_square_server]: Executing (fake) square of size 50.0  
[INFO] [1773117838.761678565] [turtle_square_server]: Goal succeeded.  
□
```

```
yahboom@yahboom-virtual-machine: ~  
yahboom@yahboom-virtual-machine:~$ source /opt/ros/humble/setup.bash  
yahboom@yahboom-virtual-machine:~$ source ~/S26_RobotProgramming_Eloquent_Elusor/Lab4/wojnae/roscourse_ws/install/setup.bash  
yahboom@yahboom-virtual-machine:~$ ros2 run turtle_test make_square  
[INFO] [1773117837.243013804] [square_client]: Waiting for action server..  
[INFO] [1773117837.244645085] [square_client]: Sending goal request with square_size=50.0..  
[INFO] [1773117837.251526780] [square_client]: Goal accepted  
[INFO] [1773117837.253884576] [square_client]: Received feedback: geometry_msgs.msg.Pose(position=geometry_msgs.msg.Point(x=10.0, y=0.0, z=0.0), orientation=geometry_msgs.msg.Quaternion(x=0.0, y=0.0, z=0.0, w=1.0))  
[INFO] [1773117837.555738011] [square_client]: Received feedback: geometry_msgs.msg.Pose(position=geometry_msgs.msg.Point(x=20.0, y=0.0, z=0.0), orientation=geometry_msgs.msg.Quaternion(x=0.0, y=0.0, z=0.0, w=1.0))  
[INFO] [1773117837.857919287] [square_client]: Received feedback: geometry_msgs.msg.Pose(position=geometry_msgs.msg.Point(x=30.0, y=0.0, z=0.0), orientation=geometry_msgs.msg.Quaternion(x=0.0, y=0.0, z=0.0, w=1.0))  
[INFO] [1773117838.159408818] [square_client]: Received feedback: geometry_msgs.msg.Pose(position=geometry_msgs.msg.Point(x=40.0, y=0.0, z=0.0), orientation=geometry_msgs.msg.Quaternion(x=0.0, y=0.0, z=0.0, w=1.0))  
[INFO] [1773117838.461473917] [square_client]: Received feedback: geometry_msgs.msg.Pose(position=geometry_msgs.msg.Point(x=50.0, y=0.0, z=0.0), orientation=geometry_msgs.msg.Quaternion(x=0.0, y=0.0, z=0.0, w=1.0))  
[INFO] [1773117838.765645609] [square_client]: Goal succeeded! Result: geometry_msgs.msg.Pose(position=geometry_msgs.msg.Point(x=50.0, y=0.0, z=0.0), orientation=geometry_msgs.msg.Quaternion(x=0.0, y=0.0, z=0.0, w=1.0))  
yahboom@yahboom-virtual-machine:~$
```

Test with non-default square size:

```
yahboom@yahboom-virtual-machine: ~  
yahboom@yahboom-virtual-machine:~$ source /opt/ros/humble/setup.bash  
yahboom@yahboom-virtual-machine:~$  
source ~/S26_RobotProgramming_Eloquent_Elusor/Lab4/wojnae/roscourse_ws/install/setup.bash  
yahboom@yahboom-virtual-machine:~$ ros2 run turtle_test turtle  
[INFO] [1773117811.502597234] [turtle_square_server]: MakeSquare action server ready.  
[INFO] [1773117837.247764716] [turtle_square_server]: Received goal: square_size=50.0  
[INFO] [1773117837.250685272] [turtle_square_server]: Executing (fake) square of size 50.0  
[INFO] [1773117838.761678565] [turtle_square_server]: Goal succeeded.  
[INFO] [1773118009.502725586] [turtle_square_server]: Received goal: square_size=200.0  
[INFO] [1773118009.505032792] [turtle_square_server]: Executing (fake) square of size 200.0  
[INFO] [1773118011.017222795] [turtle_square_server]: Goal succeeded.
```

```
yahboom@yahboom-virtual-machine: ~  
_msgs.msg.Pose(position=geometry_msgs.msg.Point(x=50.0, y=0.0, z=0.0), orientation=geometry_msgs.msg.Quaternion(x=0.0, y=0.0, z=0.0, w=1.0))  
[INFO] [1773117838.765645609] [square_client]: Goal succeeded! Result: geometry_msgs.msg.Pose(position=geometry_msgs.msg.Point(x=50.0, y=0.0, z=0.0), orientation=geometry_msgs.msg.Quaternion(x=0.0, y=0.0, z=0.0, w=1.0))  
yahboom@yahboom-virtual-machine:~$ ros2 run turtle_test make_square --ros-args --param square_size:=200.0  
[INFO] [1773118009.497659283] [square_client]: Waiting for action server..  
[INFO] [1773118009.499185325] [square_client]: Sending goal request with square_size=200.0..  
[INFO] [1773118009.505054532] [square_client]: Goal accepted  
[INFO] [1773118009.507617428] [square_client]: Received feedback: geometry_msgs.msg.Pose(position=geometry_msgs.msg.Point(x=40.0, y=0.0, z=0.0), orientation=geometry_msgs.msg.Quaternion(x=0.0, y=0.0, z=0.0, w=1.0))  
[INFO] [1773118009.811127564] [square_client]: Received feedback: geometry_msgs.msg.Pose(position=geometry_msgs.msg.Point(x=80.0, y=0.0, z=0.0), orientation=geometry_msgs.msg.Quaternion(x=0.0, y=0.0, z=0.0, w=1.0))  
[INFO] [1773118010.113093750] [square_client]: Received feedback: geometry_msgs.msg.Pose(position=geometry_msgs.msg.Point(x=120.0, y=0.0, z=0.0), orientation=geometry_msgs.msg.Quaternion(x=0.0, y=0.0, z=0.0, w=1.0))  
[INFO] [1773118010.415370406] [square_client]: Received feedback: geometry_msgs.msg.Pose(position=geometry_msgs.msg.Point(x=160.0, y=0.0, z=0.0), orientation=geometry_msgs.msg.Quaternion(x=0.0, y=0.0, z=0.0, w=1.0))  
[INFO] [1773118010.716518486] [square_client]: Received feedback: geometry_msgs.msg.Pose(position=geometry_msgs.msg.Point(x=200.0, y=0.0, z=0.0), orientation=geometry_msgs.msg.Quaternion(x=0.0, y=0.0, z=0.0, w=1.0))  
[INFO] [1773118011.022250027] [square_client]: Goal succeeded! Result: geometry_msgs.msg.Pose(position=geometry_msgs.msg.Point(x=200.0, y=0.0, z=0.0), orientation=geometry_msgs.msg.Quaternion(x=0.0, y=0.0, z=0.0, w=1.0))  
yahboom@yahboom-virtual-machine:~$
```

Pre-Lab 4 - Part 2

Modified launch file:

```
from launch import LaunchDescription
from launch_ros.actions import Node
from launch.actions import DeclareLaunchArgument
from launch.substitutions import LaunchConfiguration, TextSubstitution

def generate_launch_description():
    name_arg = DeclareLaunchArgument(
        'my_parameter',
        default_value=TextSubstitution(text='default')
    )

    talk1 = Node(
        package='python_parameters',
        namespace='group6',
        executable='minimal_param_node',
        name='Hiro',
        parameters=[{'my_parameter': 'Hiro'}]
    )

    talk2 = Node(
        package='python_parameters',
        namespace='group6',
        executable='minimal_param_node',
        name='Baymax',
        parameters=[{'my_parameter': 'Baymax'}],
        arguments=["--ros-args", "--log-level", "WARN"]
    )

    talk3 = Node(
        package='python_parameters',
        namespace='group0',
        executable='minimal_param_node',
        name='NA',
        parameters=[{
            'my_parameter': LaunchConfiguration('my_parameter'),
            'wait_time': 2.5
        }]
    )

    return LaunchDescription([
        name_arg,
        talk1, talk2, talk3
    ])
```

Test initial launch:

```
yahboom@yahboom-virtual-machine: ~  
nah~/group0 --params-etc /tmp/launch-params_app11cc j.  
yahboom@yahboom-virtual-machine:~$ ros2 launch python_parameters run_multi_launch.py  
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-03-10-  
22-27-05-232088-yahboom-virtual-machine-4251  
[INFO] [launch]: Default logging verbosity is set to INFO  
[INFO] [minimal_param_node-1]: process started with pid [4252]  
[INFO] [minimal_param_node-2]: process started with pid [4254]  
[INFO] [minimal_param_node-3]: process started with pid [4256]  
[minimal_param_node-1] [INFO] [1773152827.172500848] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-1] [INFO] [1773152828.138114009] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-3] [INFO] [1773152828.677998710] [group0.NA]: Hello NA!  
[minimal_param_node-1] [INFO] [1773152829.137889246] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-1] [INFO] [1773152830.137979208] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-1] [INFO] [1773152831.138554083] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-3] [INFO] [1773152831.150230568] [group0.NA]: Hello NA!  
[minimal_param_node-1] [INFO] [1773152832.137407292] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-1] [INFO] [1773152833.137975671] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-3] [INFO] [1773152833.650995114] [group0.NA]: Hello NA!  
[minimal_param_node-1] [INFO] [1773152834.138188971] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-1] [INFO] [1773152835.137892724] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-1] [INFO] [1773152836.137859375] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-3] [INFO] [1773152836.150718426] [group0.NA]: Hello NA!  
[minimal_param_node-1] [INFO] [1773152837.139383526] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-1] [INFO] [1773152838.138420668] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-3] [INFO] [1773152838.650736997] [group0.NA]: Hello NA!  
[minimal_param_node-1] [INFO] [1773152839.138858409] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-1] [INFO] [1773152840.138873469] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-1] [INFO] [1773152841.138152950] [group6.Hiro]: Hello Hiro!  
[minimal_param_node-3] [INFO] [1773152841.150634553] [group0.NA]: Hello NA!
```

```
yahboom@yahboom-virtual-machine: ~/S26_RobotProgramming_El...  
yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab4/wojnae/ro  
source_ws$ ros2 node list  
/group0/NA  
/group6/Baymax  
/group6/Hiro  
yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab4/wojnae/ro  
source_ws$
```

Test modified launch file with argument:

```
yahboom@yahboom-virtual-machine: ~/S26_RobotProgramm...
yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab4/wojnae/roscourse_ws$ ros2 launch python_parameters run_multi_launch.py my_parameter:=GoGo
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-03-10-22-45-21-702162-yahboom-virtual-machine-5471
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [minimal_param_node-1]: process started with pid [5472]
[INFO] [minimal_param_node-2]: process started with pid [5474]
[INFO] [minimal_param_node-3]: process started with pid [5476]
[minimal_param_node-1] [INFO] [1773153923.626405460] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773153924.599583432] [group6.Hiro]: Hello Hiro!
[minimal_param_node-3] [INFO] [1773153925.104156734] [group0.NA]: Hello GoGo!
[minimal_param_node-1] [INFO] [1773153925.599505619] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773153926.598686387] [group6.Hiro]: Hello Hiro!
[minimal_param_node-3] [INFO] [1773153927.575560949] [group0.NA]: Hello GoGo!
[minimal_param_node-1] [INFO] [1773153927.599860962] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773153928.599111698] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773153929.599210913] [group6.Hiro]: Hello Hiro!
[minimal_param_node-3] [INFO] [1773153930.075757052] [group0.NA]: Hello GoGo!
[minimal_param_node-1] [INFO] [1773153930.598792519] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773153931.599355660] [group6.Hiro]: Hello Hiro!
[minimal_param_node-3] [INFO] [1773153932.575773077] [group0.NA]: Hello GoGo!
[minimal_param_node-1] [INFO] [1773153932.599849280] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773153933.599129952] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773153934.599079017] [group6.Hiro]: Hello Hiro!
[minimal_param_node-3] [INFO] [1773153935.075706625] [group0.NA]: Hello GoGo!
```

```
yahboom@yahboom-virtual-machine: ~/S26_R...
yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab4/wojnae/roscourse_ws$ ros2 node list
/group0/NA
/group6/Baymax
/group6/Hiro
yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab4/wojnae/roscourse_ws$
```